Woodward Academy Model UN
Background Guide
General Assembly
Generative AI and The Global Water Crisis

Background:

Generative AI is artificial intelligence that creates new content like text, images, code, or audio when you give it a prompt. This is different from older, more traditional AI, which is usually built to do one specific job, like flagging fraud or giving directions, using data trained just for that job. Generative models learn from huge, mixed datasets and then produce new content that looks similar to what they learned from, without directly copying it. The technology behind this has existed for over a decade, but it did not really take off until 2017, when a new model design called the transformer made it possible to build much bigger and more powerful systems. Once OpenAI released ChatGPT to the public in late 2022, generative AI went from something researchers talked about to something almost everyone was using within months. Since then, companies have been racing to build bigger and more advanced models. This committee has to weigh what generative AI is actually giving the world against what it costs, especially when it comes to water, and figure out what the international community should do about it.

Benefits and risks of generative AI:

A lot of the excitement around generative AI comes down to money. McKinsey estimates it could add 2.6 to 4.4 trillion dollars a year to the global economy just through faster customer service, marketing, coding, and research. Not everyone agrees on how big of a deal this really is, though. MIT economist Daron Acemoglu thinks AI will probably raise U.S. productivity by less than one percent over the next ten years, which he calls a real but small gain, and he warns that most of that gain could end up going to companies and investors rather than workers. AI's scientific benefits are less controversial. Researchers at the Max Planck Institute, for example, used AI to help design a quantum physics experiment that scientists had been stuck on for weeks, and the National Academies point to work like this as a real benefit of the technology.

Generative AI also comes with risks that have nothing to do with water. As more AI-generated text and images spread across the internet, newer AI models are increasingly trained on content that older AI models made instead of content made by actual people. Researchers call this model collapse, and it slowly makes models less accurate and less reliable over time. AI is also trained on huge amounts of copyrighted material scraped from the internet, and that has led to lawsuits. The New York Times sued OpenAI and Microsoft in December 2023, claiming they used its

articles to train AI models without permission. Courts have ruled that AI-generated content itself cannot be copyrighted, but whether it is legal to train a model on copyrighted material without a license is still an open question. On top of that, some investors and journalists have started comparing how fast AI company valuations are rising to the dot-com bubble of 2000, even though today's biggest AI companies actually make real money, unlike most companies from that earlier crash.

Why generative AI needs water, and how much:

Running AI models takes huge clusters of powerful computer chips that work nonstop, and that generates a lot of heat. Most data centers cool this down using evaporative cooling, where water absorbs heat and evaporates out of cooling towers, kind of like how sweat cools down the human body. There is also a hidden cost: the electricity that powers these data centers usually comes from power plants that use water themselves to cool their turbines. Worldwide, AI is expected to need somewhere between 4.2 and 6.6 billion cubic meters of water by 2027, which is more than an entire country like Denmark uses in a year. That number does not spread out evenly, though. It builds up wherever a data center is actually located, so one facility can put real strain on a local water supply even if the global numbers still look manageable overall.

The United Nations and the global water crisis:

AI's growing need for water is hitting a world that already does not have enough freshwater to go around. More than two billion people still do not have access to safely managed drinking water. The UN has known this was a problem for a while now. Back in 2016, the General Assembly declared 2018 to 2028 the International Decade for Action on "Water for Sustainable Development," meant to help the world reach Sustainable Development Goal 6, which is clean water and sanitation for everyone. A progress check partway through that decade found the world was way behind schedule and called on governments to move much faster. AI policy at the UN has developed almost completely separately from all of this. The General Assembly's first big resolution on AI, passed in March 2024, focused on human rights and safety, but did not mention water at all, and right now there is still no global rule requiring AI companies to track or limit how much water they use.

Regional Perspectives:

Frontier AI Powers: The United States and China host most of the world's most advanced AI infrastructure, and both countries have mostly treated fast AI growth as a matter of staying competitive. Neither has really regulated data center water use at a national level, leaving most of that up to state or local governments.

Water-Stressed Host Nations: Companies have started building more data centers in countries where land and construction are cheaper, and some of those countries are already dealing with

drought. In Chile, for instance, courts blocked a planned Google data center near Santiago after ruling that the company had not properly accounted for how much it would strain the local water supply.

Developing and Lower-Income Economies: There is also a divide over who actually benefits from AI economically, separate from who hosts the physical infrastructure. Stanford's AI Index found that 83 percent of people in China see AI as more helpful than harmful, compared to only 39 percent in the United States, which shows how differently people view AI depending on where they live and what their economy looks like.

Technology Companies: Microsoft, Google, and Amazon have all promised to become "water positive" by 2030, meaning they plan to put back more water globally than they use. Critics point out that replacing water in one part of the world does not really help a community whose local water supply is already strained right now, like in Chile.

Moving forward, this committee has to decide how the international community should handle a technology that is reshaping economies, helping scientists make real discoveries, and putting pressure on a resource that billions of people already struggle to access. Should AI companies be required to track and report how much water they use, the same way many are required to track carbon emissions? Should governments have to consider water specifically before approving new data center projects? Since experts still disagree on how big AI's economic benefits will actually turn out to be, how much water risk is worth accepting for benefits that are still uncertain? Delegates should also think about whether the wealthy countries pushing AI forward and the developing countries increasingly hosting its infrastructure can actually work together on this, or whether their interests are just too different.

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